

Beautified Is Not Forged: Separating Beautification from Face Forgery

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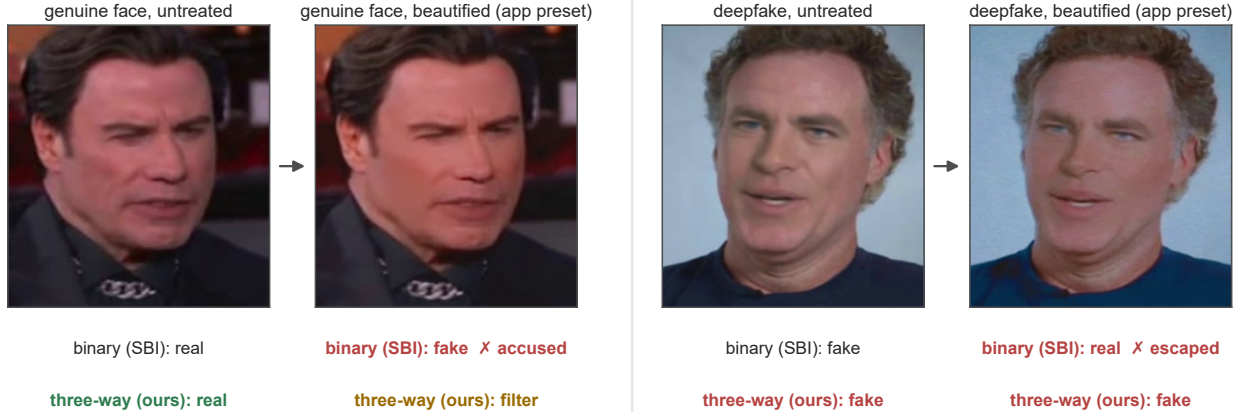


Fig. 1. Celeb-DF-B frames of a genuine video and of a deepfake, before and after the same app beautification. The official SBI checkpoint, at its 5%-FPR threshold on untreated genuine frames, accuses the beautified genuine face and lets the beautified deepfake pass. Ours calls the first *filter* and the second *fake*. Frames were chosen where SBI errs after beautification; rates over the benchmark are in Table III.

Abstract—Face-forgery detectors decide between real and fake, while many genuine faces have been beautified by a camera app or a retouching service. We measure what this does to thirteen published detectors, including two CLIP-based detectors from 2025. When genuine and forged FaceForensics++ frames receive the same beautification, keeping the escape of beautified forgeries under 5% makes every detector accuse at least 14.5% of beautified genuine faces. On Celeb-DF-B, app beautification raises the share of deepfakes called real by 3.4 to 35.5 percentage points for the eleven detectors that detect its deepfakes above chance. We add a third class, *filter*, for edits that change how a face looks but not who it shows or what the person does; with everything else in training held fixed, it removes this effect. A filter class learnt from scripted operations, however, does not recognise commercial retouching: FF++-trained models separate retouched from untouched faces at chance. We train the three-way detector on the renders of two commercial retouching services, split by optical flow and frequency into single-operation components with exact edit footprints, and test it on a third service. A CLIP ViT-L/14 model with LoRA trained this way calls 0.1% of the unseen service’s retouched faces fake, matches Effort and Forensics Adapter on Celeb-DF-v2 and DFD (video-level AUC 0.954 and 0.983), and is not affected by beautification once re-encoding is accounted for. Its evidence head names regions whose restoration removes 91% of its detections on held-out part-level forgeries. Telling the service’s lightly retouched faces from their originals remains difficult (70% balanced accuracy).

I. INTRODUCTION

Face-forgery detectors are meant for content moderation, remote identity checks and fact-checking. In these settings a genuine face called fake accuses a real person, and a forgery called real is let through. Many of the genuine faces

such a detector receives have been retouched: retouching is common on social media and photo-sharing sites and even on identification cards [1], and skin smoothing, whitening, eye enlargement and face slimming are default options in camera apps and are sold by retouching APIs. Beautification edits the pixels of the face, which is what a forgery detector learns to inspect, but it changes neither who is shown nor what the person does.

We measure both errors on thirteen published detectors. On FaceForensics++ (FF++) [2] frames beautified in the same way whether genuine or forged, no threshold of any detector lets at most 5% of the beautified forgeries pass while calling fewer than 14.5% of the beautified genuine faces fake. On a commercial retouching service that none of them was trained on, at a threshold that accuses 5% of the untouched faces, the eleven detectors trained on FF++ accuse 36–63% of the most strongly smoothed ones. Beautification also works as an evasion. On Celeb-DF-B [3], once the effect of re-encoding is removed, app beautification lets 21.8, 11.0 and 15.3 percentage points more deepfakes pass SBI [4], Effort [5] and Forensics Adapter [6]. Libourel et al. [3], who released Celeb-DF-B, measured drops of 0.15–0.16 in the video-level AUC of three detectors and concluded that “the use of a dedicated filter detection method is strongly advisable”. Saeed et al. [7] show that enhancement lets deepfakes evade detectors, and Concas et al. [8] that beauty filters raise the error of deepfake and morphing-attack detectors.

The two errors come from the label space. A binary detector

has to call a beautified genuine face real or fake. Calling it fake is a false accusation. Calling it real teaches the detector that beautification is a sign of authenticity, and a beautified deepfake then passes as real (Fig. 1). A threshold only trades one error for the other, and for none of the thirteen detectors does any threshold reach the pair of errors of our three-way detector (Sec. V). We add a third label, *filter*, for edits that change how a face looks but not who it shows or what the person does, and define the three labels by what an edit changes rather than by the tool that made it (Sec. III-A).

The third label does not suffice by itself. Detectors trained on FF++ call blurred genuine faces fake, and a filter class trained on hand-written beautification scripts misses the retouching of commercial services. Renders of commercial services would teach it, but they apply several operations at once, and a model trained on them learns which operations co-occur. We therefore split each render into single-operation components that are pixel-aligned with the original, which gives both a label and an exact footprint for every edit.

Our contributions are:

- **A measurement of the problem.** Threshold sweeps of thirteen detectors on a paired FF++ corpus and on Celeb-DF-B, a same-pipeline replication that isolates the label space, and 2,210 before/after pairs from a commercial service as a test of retouching recognition.
- **Supervision from commercial renders.** A decomposition of multi-operation renders into single-operation components (Alg. 1), used with degradation applied equally to all classes and with part-level forgeries to train one three-way detector.
- **Evidence with a faithfulness test.** An evidence head trained on exact edit footprints, and a restoration test that checks whether the region it names is sufficient and necessary for the decision.

The three-class formulation itself is not new. ForgeryNet [9] has a three-way task in which identity-preserving edits count as forgeries, and Bobulski and Kubanek [10] classify real, photo-edited and generated faces. We assign beautification to a genuine outcome and train with commercial renders.

II. RELATED WORK

Beautification and forgery detection. The studies cited in the introduction [3], [7], [8] evaluate existing detectors; none of them trains a detector that removes the effect.

Retouching detection. RetouchingFFHQ [11] labels four commercial operations at three levels and reports cross-vendor recall. Diffusion models restore the untouched face [12], [13], a level classifier can guide the reversal [14], and predicted warp fields undo Photoshop liquify [15]. These methods have no forgery class.

Generalisable forgery detection. Face X-ray [16] and SBI [4] synthesise blending artefacts from real images, and UCF [17] separates common forgery features from content and method-specific ones. Effort [5] and Forensics Adapter [6] adapt CLIP ViT-L/14 with few trainable parameters; they

TABLE I
LABELS OF CASES NEAR THE CLASS BOUNDARY. LAST COLUMN: WHETHER THE CASE OCCURS IN OUR TRAINING OR TEST DATA.

Case	Label	In our data
Camera processing, JPEG, video coding	REAL	all sets
Make-up worn at capture	REAL	incidental
Beautification by an app or a service, neural or not	FILTER	scripted, 3 services, app presets
Reenactment by another person's performance	FAKE	Face2Face, NeuralTextures
A part from another face, or inpainted	FAKE	part-level forgeries
Face swap	FAKE	FF++, Celeb-DF, DFD
Beautified forgery	FAKE	FF++, Celeb-DF-B
Expression editing; synthesised face	FAKE	not tested

are the most recent detectors in our comparison. DeepfakeBench [18] provides common checkpoints for many detectors.

Explanations for detectors. GAIN [19] trains attention maps so that erasing the attended region removes the class evidence, amortised maskers predict saliency in one pass [20], [21], and DOLOS [22] compares explanation, local-score and attention methods for localising inpainted face regions and finds local scores best. We supervise the map with exact footprints and test it by restoring pixels.

III. METHOD

A. Labels and errors

The label of a face x is $y \in \{\text{REAL}, \text{FAKE}, \text{FILTER}\}$, decided by what was changed after the capture, the photograph or video frame as recorded, including in-camera processing and compression.

- **REAL:** the capture, with no later edit of the face.
- **FILTER:** the face is still the captured person's own face at the captured moment, and only its appearance was changed: colour and tone, skin texture, or a smooth re-shaping of the person's own features (smoothing, whitening, eye enlargement, face slimming, colour grading). Identity, expression, pose and speech are those of the capture.
- **FAKE:** the image shows a person or an action that was not captured: an identity swap, a reenactment (the captured identity driven by another performance), a synthesised face, or a facial part replaced by another person's part or by generated content.

An image that contains a fake edit is **FAKE** whatever else was applied, so a beautified forgery is **FAKE**. The tool does not decide the label: a retouching service that uses a neural network is **FILTER**, and a reenactment is **FAKE** although it keeps the identity. Table I lists cases near the boundary and whether our data contains them.

The detector outputs class probabilities $p(x)$, presence scores $\hat{s}(x) \in [0, 1]^4$ for eye, contour, tone and texture edits, and an evidence map $\hat{m}(x)$. The decision is $\hat{y} = \arg \max_c p_c(x)$.

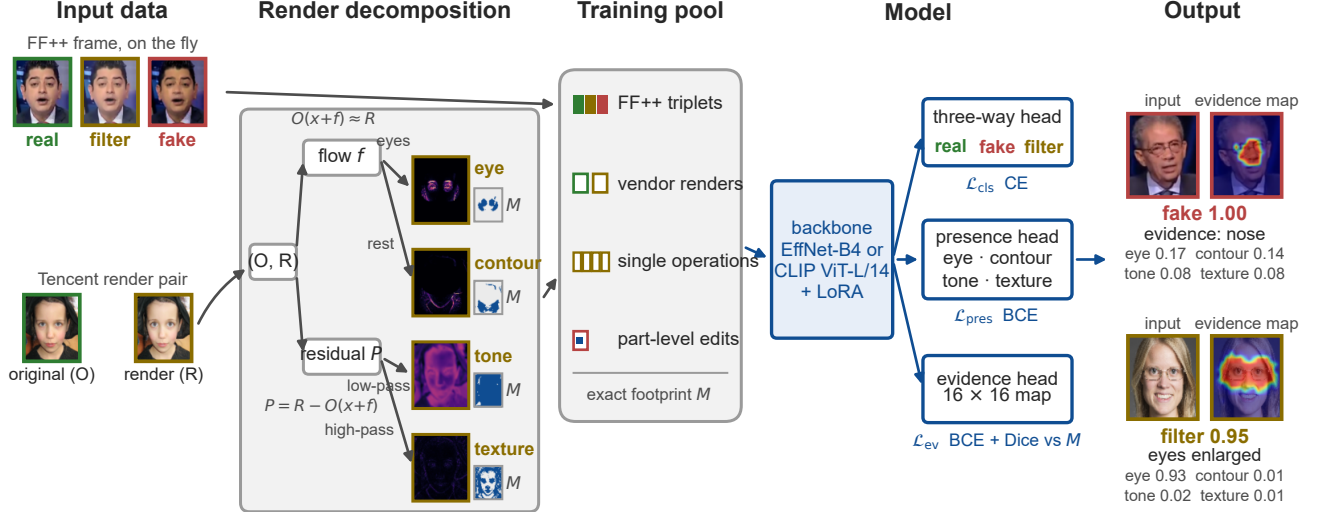


Fig. 2. Overview. Each genuine FF++ photograph yields a real / filter / fake triplet on the fly; a beautified forgery stays *fake*. Each commercial render R of an original O is split by its optical flow f and photometric residual P into four single-operation components (shown as $|\text{component} - O|$), each with an exact footprint M (Eq. 1, Alg. 1). One backbone feeds a three-way head, an operation-presence head and an evidence head trained against M (Eq. 4). Right: outputs of Ours on a held-out part-level forgery and on a render from the unseen service.

Two error rates describe beautification. FA is the share of beautified genuine faces with $\hat{y} = \text{FAKE}$ and MISS the share of beautified forgeries with $\hat{y} = \text{REAL}$; a beautified forgery sent to FILTER counts in neither. A binary detector with score s traces a curve $(\text{FA}(t), \text{MISS}(t))$ over thresholds t and *dominates* a three-way detector with errors $(\text{FA}^*, \text{MISS}^*)$ if some t gives $\text{FA}(t) \leq \text{FA}^*$ and $\text{MISS}(t) \leq \text{MISS}^*$. We sweep every threshold on the test set, which favours the binary detectors.

B. Supervision from paired originals

Every edited image x in our training pool has a pixel-aligned original x_0 , and the pair defines where the edit is:

$$M(x) = \text{dilate}_\rho(\mathbf{1}[\|x - x_0\|_\infty > \tau]), \quad (1)$$

with $\tau = 6$ on 8-bit intensities and ρ the feather width of the edit. Observed FF++ forgeries have no aligned original and get no footprint. The pool has four sources (Fig. 2).

FF++ triplets. FF++ c23 under the official 720/140/140 video split. For each genuine training photograph x_0 a step draws a REAL item (x_0), a FILTER item (an on-the-fly beautification of x_0 by one of four landmark and skin operations at a random strength or by one of 14 photometric filters [23], otherwise an observed filter frame) and a FAKE item (a self-blend of x_0 [4], an observed forgery from the four FF++ methods, or a part-level forgery). A fake is beautified on top with probability 0.5. The same random blur or downscale is applied to all three items, so that sharpness does not predict the class.

Commercial renders. 7,831 FFHQ photographs [24] (indices 60002–69999) enter REAL and serve as self-blend bases, so that a clean still photograph is not a class cue. Their renders by two retouching APIs from RetouchingFFHQ [11], Tencent

(7,730 renders at levels 30/60/90) and Megvii (13,139), enter FILTER. Both services apply eye enlargement, face lifting, whitening and smoothing together, and a model trained on the renders learns to rely on the texture change (Sec. V-E). We split each render R of an original O into single-operation components in O 's geometry (Alg. 1). Dense optical flow [25] on locally normalised luminance gives f with $O(\mathbf{u} + f(\mathbf{u})) \approx R(\mathbf{u})$, and the photometric residual in O 's geometry is

$$P(\mathbf{u}) = R(\mathbf{u} - f^{-1}(\mathbf{u})) - O(\mathbf{u}). \quad (2)$$

With E the eye-region mask, $\mathcal{W}(O; g)$ the warp of O by a flow g and G_σ a Gaussian ($\sigma = 6$ px at 512 px), the components are

$$\begin{aligned} C_{\text{eye}} &= \mathcal{W}(O; E \odot f), & C_{\text{contour}} &= \mathcal{W}(O; (1 - E) \odot f), \\ C_{\text{tone}} &= O + G_\sigma * P, & C_{\text{texture}} &= O + P - G_\sigma * P. \end{aligned} \quad (3)$$

Applying the four edits in sequence reproduces R at 108 dB PSNR. Each component is a FILTER item whose presence target is its operation. Components whose measured edit (below) falls under the noise floor of the measurement are dropped, leaving 50,112 of 73,952. Parametric single operations, with strengths drawn log-uniformly from the range measured on the two services, complete the filter pool. The third service, Alibaba (FFHQ indices 17001–19999, one operation per render), is used only for testing, and the content keys of its 28,729 images and of the 7,831 training bases are disjoint.

Part-level forgeries. We build 61,510 training forgeries and a held-out test set on FF++ frames by transplanting the eyes, nose or mouth of another identity (landmark-aligned warp, colour matching, feathering) or by regenerating the part with Stable Diffusion 1.5 inpainting. Pixels outside the feathered

Algorithm 1 Single-operation components of a commercial render

Require: original O , render R , eye mask E , σ , noise floors

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1:  $f \leftarrow \text{DIS}(\text{LCN}(O), \text{LCN}(R)) \quad \triangleright \text{normalised luminance}$ 
2:  $P \leftarrow R \circ (\text{id} - f^{-1}) - O \quad \triangleright \text{Eq. 2}$ 
3:  $C_{\text{eye}} \leftarrow \mathcal{W}(O; E \odot f); \quad C_{\text{contour}} \leftarrow \mathcal{W}(O; (1 - E) \odot f)$ 
4:  $C_{\text{tone}} \leftarrow O + G_{\sigma} * P; \quad C_{\text{texture}} \leftarrow O + P - G_{\sigma} * P$ 
5: for  $k \in \{\text{eye, contour, tone, texture}\}$  do
6:    $q_k \leftarrow \text{Measure}_k(C_k, O)$ 
7:   if  $q_k \geq \tau_k$  then
8:     keep  $(C_k, e_k, M(C_k)) \quad \triangleright \text{Eq. 1}$ 
9:   end if
10: end for
```

mask are copied bit-exactly from x_0 , so Eq. 1 is exact. SDXL inpainting is generated for test frames only.

Edit measurements. For each aligned pair we measure $q = (r_{\text{eye}} - 1, 1 - r_{\text{jaw}}, \Delta L/20, 1 - r_{\text{hf}})$, the eye- and jaw-width ratios from re-detected landmarks, the CIELAB lightness change and the high-frequency energy ratio inside the face hull. The measurements set the noise floors τ_k . A head that regressed q did not pass validation (supplement).

C. Model and objective

A backbone ϕ feeds three heads. For Ours it is a frozen CLIP ViT-L/14 [26] with rank-16 LoRA [27] on every attention projection (3.2M trained parameters of 306M); for Ours-lite it is EfficientNet-B4 [28] (17.6M). The three-way head and the presence head read the pooled feature, and the evidence head $\hat{m} = \sigma(g(\phi_{\text{sp}}(x)))$ is a 1×1 convolution on the spatial tokens (16×16 for CLIP, 12×12 for EfficientNet-B4). The loss on one item is

$$\mathcal{L} = \text{CE}(p, y) + \lambda_s \mathbf{1}[y = \text{FILTER}] \text{BCE}(\hat{s}, s) + \lambda_m \mathbf{1}[M \text{ exact}] (\text{BCE}(\hat{m}, M) + \text{Dice}(\hat{m}, M)), \quad (4)$$

with $\lambda_s = \lambda_m = 1$. Real items have an empty footprint and observed forgeries no footprint term. The classifier does not read $\hat{m}$, so whether the map explains the decision has to be tested.

Faithfulness test. For a detected part-level forgery with footprint M , let $\hat{A}$ be the $|M|$ highest cells of $\hat{m}$, dilated by the feather width, and let $x_A = x \odot (1 - A) + x_0 \odot A$ restore a region A to the original. We report the flip rate $\Pr[\arg \max p(x_{\hat{A}}) \neq \text{FAKE}]$ and the same rate for the complement $1 - \hat{A}$, for the true footprint M (the ceiling), for a centred region and for a random region of equal area, together with the IoU of $\hat{m}$ and M on a 7×7 grid (IoU_7). A faithful map flips as often as the true footprint and its complement does not flip.

Training. CLIP: 224px, AdamW at 10^{-4} with warm-up and cosine decay, 8 epochs. EfficientNet-B4 (AdvProp weights [29]): 380px, SGD at 10^{-3} with SAM [30], 12 epochs. A batch holds 16 genuine bases (48 items); fakes are drawn as self-blends (0.4), observed forgeries (0.3) or

part-level forgeries (0.3). Checkpoints are selected on FF++ validation macro-F1 plus F1 on held-out Tencent and Megvii bases, which picks epoch 4 of 8 for Ours and epoch 12 of 12 for Ours-lite (curves in the supplement). No test set is used for any choice.

IV. EXPERIMENTAL SETUP

Partitions. Table II gives the training, validation and test data of each source. FF++ follows the official video split, so no video is shared between partitions, and part-level forgeries are built on frames of the matching partition. The commercial validation data are FFHQ bases of the two training services that were held out from training; the commercial test data come from a third service.

Test sets. FF++ c23 test with 400 test forgeries under eight beautification conditions; Celeb-DF-v2 [31] (official list, 518 videos, 16,059 frames); DFD [32] (332 held-out videos, 4,221 frames); Celeb-DF-B in the release provided by its authors, with 232 genuine and 232 synthesis videos, each untreated, beautified by one of four app presets and re-encoded without a filter (43,363 frames); and the Alibaba pairs (four operations at three levels). Video sets use 32 evenly spaced frames per video.

Metrics. Frame-level AUC and video-level AUC (mean of the 32 frame scores), with 95% video-cluster bootstrap intervals; model differences use a paired bootstrap on the same frames. For commercial retouching, balanced accuracy averages the share of renders called FILTER and the share of originals not called FILTER.

Baselines. Thirteen detectors with their released weights: eight DeepfakeBench checkpoints (Xception, EfficientNet-B4, F3Net, SPSL, UCF, RECCE, CORE, SRM), the official SBI, Effort and Forensics Adapter, UnivFD [33] and NPR [34]. All are scored on our frames with our crops. For Xception, EfficientNet-B4 and UCF, whose frame-level Celeb-DF-v2 AUCs are published for this protocol, our scores are within 0.02 of them (supplement).

Protocol. Target values for every model were written down before training and are reported whether met or not. Each model reads the Alibaba set once.

V. RESULTS

A. Training, validation and test accuracy

On FF++, macro-F1 falls from 91.8 on training frames to 87.2 on validation and 82.4 on test for Ours (91.4, 81.9 and 81.0 for Ours-lite; Table II). Most of the fall is on genuine frames, whose recall drops from 94.2 to 78.0; on test, 13.2% of them go to FILTER and 8.8% to FAKE. Fake and filter recall stay within 2.1 points of training, and part-level forgeries fall by 2.5 points over the three test editors, one of which (SDXL) training never saw. Renders of the training services are recognised on held-out bases as on training bases (100%), and 82.3% of the unseen service’s renders go to FILTER (63.8% for Ours-lite). Untouched commercial originals behave differently: Ours keeps 59.5% of its training originals REAL, close to the 55.5% on validation and 57.9% on test.

TABLE II

DATA AND ACCURACY (%) PER PARTITION. FF++ ROWS: RECALL PER CLASS. COMMERCIAL: TRAINING AND VALIDATION DATA ARE THE TWO TRAINING SERVICES (VALIDATION: HELD-OUT FFHQ BASES), TEST DATA THE UNSEEN SERVICE; ORIGINALS SHOULD BE CALLED *REAL*, RENDERERS AND SINGLE-OPERATION COMPONENTS *FILTER*. TRAINING IMAGES ARE COUNTED BEFORE ON-THE-FLY ITEMS (FILTERS, SELF-BLENDS) ARE ADDED; TRAINING ACCURACY IS MEASURED ON A RANDOM SAMPLE OF 2,000 IMAGES PER ROW (4,000 PART-LEVEL FORGERIES), VALIDATION AND TEST ACCURACY ON EVERY IMAGE.

Data	Class	Images			Ours			Ours-lite		
		train	val	test	train	val	test	train	val	test
FF++ c23	real	41,512	530	1,316	94.2	80.9	78.0	96.9	78.5	74.2
	fake	41,526	529	5,304	89.6	90.7	88.9	94.0	92.6	93.3
	filter	41,512	530	1,316	91.3	90.0	89.2	83.3	75.1	76.4
	macro-F1				91.8	87.2	82.4	91.4	81.9	81.0
Commercial	originals → real	7,015	816	2,210	59.5	55.5	57.9	89.6	79.3	81.0
	renders → filter	18,734	2,136	26,519	99.9	100.0	82.3	99.8	100.0	63.8
	components → filter	50,112	5,684	–	94.8	94.2	–	88.7	87.9	–
Part-level	forgeries → fake	61,510	–	11,844	91.8	–	89.3	93.8	–	92.5

TABLE III

FORGERY DETECTION AND BEAUTIFICATION ERRORS ON THE SAME FRAMES. AUC OF $P(\text{FAKE})$ AT FRAME AND VIDEO LEVEL; CELEB-DF-B AUC ON UNTREATED GENUINE VS. SYNTHESIS FRAMES. FA: BEAUTIFIED GENUINE FACES CALLED FAKE, MISS: BEAUTIFIED FORGERIES CALLED REAL (%). FOR BINARY DETECTORS THE FA IS THE LOWEST ONE A THRESHOLD ON THE TEST SET REACHES WHILE KEEPING MISS AT THE STATED LEVEL; Δ MISS IS THE CHANGE CAUSED BY CELEB-DF-B BEAUTIFICATION AT 5% FPR ON UNTREATED GENUINE FRAMES. THREE-WAY MODELS USE NO THRESHOLD. BOLD: BEST PER COLUMN; Δ MISS IS NOT MARKED BECAUSE ITS SMALLEST VALUES BELONG TO DETECTORS THAT BARELY DETECT CELEB-DF-B DEEPPAKES (UNIVFD, NPR).

Detector	Params (M)	Celeb-DF-v2		DFD		Celeb-DF-B	FF++ beautified	Celeb-DF-B beautified	
		frame	video	frame	video	AUC	FA at MISS $\leq$ 5	Δ MISS	FA at MISS $\leq$ 9.7
<i>Published binary detectors, official weights, our frames; FA read off the full threshold sweep</i>									
Xception	21.9	0.718	0.772	0.897	0.882	0.811	19.1	+9.9	65.7
EfficientNet-B4	19.3	0.750	0.807	0.862	0.860	0.811	19.2	+3.4	59.2
F3Net	21.9	0.708	0.750	0.852	0.811	0.798	14.5	+6.8	66.1
SPSL	21.9	0.703	0.752	0.873	0.851	0.781	19.9	+13.1	63.3
UCF	21.9	0.744	0.788	0.877	0.853	0.841	16.1	+12.4	59.6
RECCE	47.7	0.688	0.731	0.900	0.877	0.785	22.6	+11.8	68.2
CORE	21.9	0.715	0.764	0.882	0.859	0.821	19.1	+13.8	63.4
SRM	53.2	0.717	0.778	0.893	0.869	0.789	19.1	+17.0	65.4
SBI	17.6	0.817	0.872	0.918	0.876	0.875	40.4	+35.5	67.3
Effort (CLIP-L/14)	303.4	0.881	0.930	0.935	0.957	0.909	36.5	+23.7	61.2
Forensics Adapter (CLIP-L/14)	309.7	0.887	0.945	0.942	0.959	0.909	30.5	+15.5	53.6
UnivFD	427.6	–	–	–	–	0.653	80.2	–0.2	81.7
NPR	1.4	–	–	–	–	0.503	92.4	+1.6	90.5
<i>Three-way, native argmax rule, no threshold (MISS in parentheses)</i>									
Ours (CLIP ViT-L/14 + LoRA)	306.3	0.888	0.954	0.948	0.983	0.937	4.0 (0.6)	+2.9	18.4 (9.7)
Ours-lite (EfficientNet-B4)	17.6	0.848	0.910	0.915	0.908	0.899	11.7 (0.5)	+3.9	21.9 (7.3)

Its loss of specificity (Sec. V-E) is present on images it was trained on, so it is not a failure to generalise; Ours-lite keeps 89.6% of its training originals *REAL*. Curves and confusion matrices are in the supplement.

B. Threshold sweeps

On the FF++ corpus, keeping MISS at or below 5% makes the published detectors accuse between 14.5% (F3Net) and 92.4% (NPR) of beautified genuine faces (Table III, Fig. 3). Ours accuses 4.0% and misses 0.6% with no threshold. On Celeb-DF-B, reaching our MISS of 9.7% costs the binary detectors an FA of 53.6–90.5%, against 18.4% for Ours. No threshold of any published detector dominates Ours or Ours-lite on either benchmark, and the same holds for every three-way variant in the supplement that saw real forgeries in training. The variant whose fake class contains only self-blends is dominated by six of twelve detectors on FF++.

C. Escape under beautification

We fix each binary detector at 5% FPR on untreated Celeb-DF-B genuine frames and compare the share of deepfakes called real before and after beautification (Δ MISS in Table III). The share rises by 3.4 to 35.5 pp for the eleven detectors that separate untreated Celeb-DF-B deepfakes above chance, all with intervals above zero: SBI from 44.3% to 79.8%, Effort from 27.8% to 51.4%, Forensics Adapter from 31.1% to 46.5%. Celeb-DF-B re-encodes every treated video, and its compression-only videos separate the two causes. Re-encoding alone raises the escape of SBI by 13.7 pp, leaving 21.8 pp for beautification; the remainder is 11.0 pp for Effort and 15.3 pp for Forensics Adapter. Ours rises by 2.9 pp overall and by 3.4 pp under re-encoding alone, so beautification adds nothing measurable; Ours-lite adds 0.9 pp. Ours sends 36.9% of beautified genuine frames to *FILTER*. Its largest per-preset change is +7.4 pp for the warm colour grade *califor*-

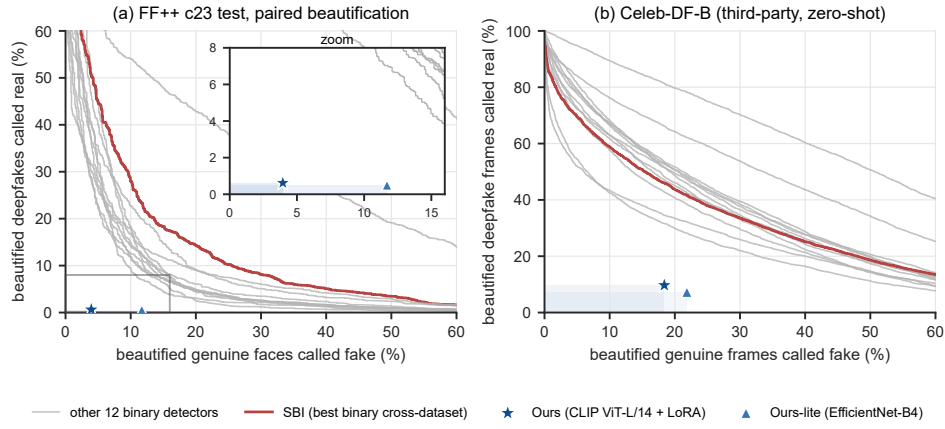


Fig. 3. Threshold sweeps. Each grey curve is one published detector over all thresholds, SBI in red. Markers are the three-way models at their native rule; the shaded box under a marker is where a binary detector would have to reach to dominate it. (a) FF++ c23 test, beautification applied to the same frames. (b) Celeb-DF-B, app beautification. No curve enters a box.

nia_dreamin (supplement).

To separate the label space from the rest of the pipeline, we retrained SBI’s binary objective with the backbone, recipe, data and frames of our FF++-only three-way model. Its escape rises by 36.8 pp [33.2, 40.8], close to the 35.5 pp of the official SBI, while the three-way model trained with the same blending changes by 0.1 pp [−1.3, 1.7].

D. Blur

On 300 genuine FF++ test frames, a Gaussian blur with $\sigma = 1.5$ makes SBI call 67% of them fake, Xception 53%, EfficientNet-B4 57% and UCF 42%, and our FF++-only three-way model 91–95%. Self-blending softens one side of the blend, and a detector trained on it learns that a soft face is a forged one. Celeb-DF genuine frames are softer than FF++ frames at the same crop size, and the FF++-only model calls 43–51% of untreated Celeb-DF-B genuine frames fake. Degrading the three items of a triplet together reduces blurred genuine frames called fake to 10% and the untreated Celeb-DF-B accusation to 28%. Ours calls 16.9% of untreated Celeb-DF-B genuine frames fake and Ours-lite 40.3%.

E. Commercial retouching

Table IV uses the Alibaba service, which applies one operation per render. Three-way models trained on FF++ alone separate retouched from untouched faces at chance (balanced accuracy 48–55% for all six variants). They move originals away from REAL and call commercial smoothing fake, up to 86% of faces at the strongest level. The eleven published detectors trained on FF++ do the same. At a threshold that calls 5% of the originals fake, chosen on these originals, they call 9.5–14.8% of retouched faces and 36–63% of the most strongly smoothed ones fake (per-detector rates in the supplement).

Ours calls 0.1% of retouched faces fake, and at the same 5% threshold 2.7% of them and 1.4% of the most strongly smoothed ones. It sends 82.3% of retouched faces to FILTER (levels 30/60/90: eye enlargement 70/79/85%, lifting

68/75/80%, whitening 72/79/85%, smoothing 96/100/100%) and gives 94.8% of renders a higher filter score than their own original. It also sends 41.9% of the untouched Alibaba originals to FILTER, and 44.4% of the originals held out from the two training services, so the loss of specificity is not limited to the unseen service; its balanced accuracy is 70.2%. Ours-lite keeps 81.0% of the originals REAL and reaches 72.6%. Our target was 75%. A threshold on $P(\text{filter})$ chosen on the training services’ validation split gives 72.4%, and the AUROC of $P(\text{filter})$ between originals and renders is 0.80–0.82 for every model trained with components, so no threshold reaches the target. The presence head recalls smoothing and whitening more often than the Megvii-to-Alibaba numbers reported by RetouchingFFHQ [11] (0.96 vs. 0.86, 0.57 vs. 0.50) and eye enlargement and lifting less often (0.21 vs. 0.52, 0.35 vs. 0.55), while marking each operation absent on 82–99% of originals, which that work does not report.

F. Forgery detection

Table III lists cross-dataset AUC with every detector scored on the same frames. Ours reaches 0.888 at frame level and 0.954 at video level on Celeb-DF-v2, 0.948 and 0.983 on DFD, and 0.937 on untreated Celeb-DF-B. The differences to the official SBI checkpoint are positive at both levels on both sets (paired intervals above zero, from [+0.006, +0.057] for DFD frames to [+0.065, +0.151] for DFD videos). The differences to Effort and Forensics Adapter are small, and all eight paired intervals contain zero, so we count Ours as level with them on these two sets. At 17.6 M parameters, Ours-lite reaches 0.910 and 0.908 at video level, above SBI on both sets. On the FF++ test split Ours recalls 78.0% of genuine, 88.9% of fake and 89.2% of filter frames; the genuine recall is below the 80% we targeted, as are the genuine (74.2%) and filter (76.4%) recall of Ours-lite.

TABLE IV

RETOUCHING BY A SERVICE NEVER SEEN IN TRAINING: 2,210 FFHQ ORIGINALS AND THEIR ALIBABA RENDERS (FOUR OPERATIONS AT THREE LEVELS, 26,519 IMAGES). PAIRED RANK: SHARE OF PAIRS IN WHICH THE RENDER GETS A HIGHER $P(\text{FILTER})$ THAN ITS OWN ORIGINAL (50 MEANS NO SIGNAL). NATIVE ARGMAX RULE. ALL OTHER VARIANTS ARE IN THE SUPPLEMENT.

Model	originals	retouched (pooled)		balanced	paired rank	paired rank at level 90	smoothing-90
	real/fake/filter	→filter	→fake			eye / white / lift / smooth	→fake
<i>FF++ c23 training only (zero-shot)</i>							
three-way, observed fakes (SUP)	36.8/5.7/57.5	66.5	9.1	54.5	79.8	97 / 89 / 79 / 59	27.3
three-way, part edits + evidence head	16.7/55.9/27.4	28.6	59.4	50.6	60.9	89 / 78 / 60 / 20	85.8
<i>Trained with two vendors' renders and their single-operation components; Alibaba never seen (ours)</i>							
Ours (CLIP ViT-L/14 + LoRA)	57.9/0.3/41.9	82.3	0.1	70.2	94.8	98 / 94 / 95 / 100	0.0
Ours-lite (EfficientNet-B4)	81.0/0.4/18.6	63.8	0.3	72.6	93.8	99 / 91 / 93 / 99	0.4

TABLE V

EXPLANATIONS ON HELD-OUT PART-LEVEL FORGERIES (FF++ TEST FRAMES, DETECTED ITEMS ONLY). IOU₇: IOU WITH THE EXACT FOOTPRINT ON A 7×7 GRID. FLIP: SHARE OF ITEMS THAT LEAVE *fake* WHEN THE NAMED REGION, OF THE SAME AREA AS THE FOOTPRINT, IS RESTORED TO THE ORIGINAL. CEIL.: THE SAME FOR THE TRUE FOOTPRINT. COMPL.: THE SAME FOR THE COMPLEMENT. SDXL WAS NEVER USED IN TRAINING. FULL TABLE IN THE SUPPLEMENT.

Map	donor transplant				SD-1.5 inpaint				SDXL inpaint (unseen editor)			
	IoU ₇	flip	ceil.	compl.	IoU ₇	flip	ceil.	compl.	IoU ₇	flip	ceil.	compl.
centre prior / random region (last col.)	0.310	38.6	–	17.0	0.249	40.9	–	17.9	0.255	40.2	–	17.7
Score-CAM, same backbone, no head	0.408	56.5	85.8	5.9	0.334	52.4	85.2	15.4	0.341	52.4	85.9	17.1
head, vendor data, no part edits (CLIP)	0.479	78.8	89.4	2.5	0.376	53.9	69.1	14.4	0.445	68.9	78.3	8.4
head, part edits, FF++ only (EffNet-B4)	0.723	84.0	84.7	0.4	0.752	82.6	83.2	0.7	0.746	82.9	83.8	0.7
Ours-lite (EffNet-B4)	0.804	87.8	87.8	0.1	0.820	87.4	87.6	0.4	0.812	86.9	87.2	0.4
Ours (CLIP ViT-L/14 + LoRA)	0.832	91.7	91.9	0.1	0.845	91.5	91.6	0.2	0.827	90.9	91.2	0.5

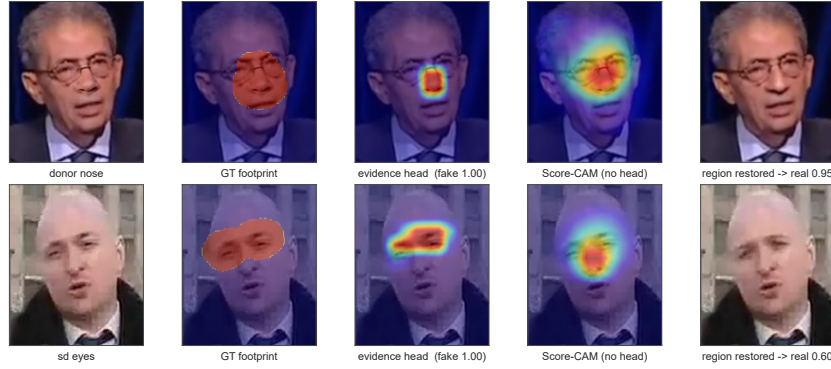


Fig. 4. Held-out part-level forgeries: input, exact footprint, evidence map with the three-way decision, Score-CAM on the same backbone without the head, and the result of restoring the named region to the original.

G. Localisation of part-level forgeries

On the three held-out editors the evidence head of Ours reaches IoU₇ 0.83–0.85 with the true footprint, and restoring the region it names removes 90.9–91.7% of its detections, against 91.2–91.9% when the true footprint is restored (Table V, Fig. 4). Restoring the complement removes at most 0.5%, a centred region 42–52% and a random region of equal area 11–15%. The SDXL editor, not seen in training, gives 0.827 and 90.9% against a ceiling of 91.2%. Ours-lite reaches 0.80–0.82 and 86.9–87.8% against 87.2–87.8%. The named region is therefore sufficient and necessary for the model’s own decision on these forgeries. Post-hoc maps on a backbone without the head do only moderately better than a centred

region (IoU₇ 0.26–0.41 and flips of 45–57%, against 0.25–0.31 and 39–41%). The footprint supervision carries the result: the same CLIP model trained without part-level forgeries reaches 0.38–0.48 and 54–79%, and an EfficientNet-B4 model trained on FF++ with part-level forgeries but without the retouching data already reaches 0.72–0.75 and 83–84%. For whole-face FF++ forgeries no method finds a small sufficient region; restoring 15% of the face removes 22–25% of detections for every method and a random region 14–21%.

H. Contribution of each training signal

The supplement varies one signal at a time. The renders alone already stop the accusation of commercial retouching (0.6% fake) but send single operations other than smoothing

to FILTER only 10–46% of the time (balanced 66.1%). The single-operation components raise balanced accuracy to 70.9–72.7% over two seeds. The CLIP backbone adds 0.05 frame AUC on Celeb-DF-v2 at equal data (73.4–74.6% balanced over two seeds), and the part-level forgeries add the evidence head and 0.005–0.010 on Celeb-DF-v2. On the CLIP backbone they also raise the share of held-out training-service originals called FILTER from 25–28% to 44% and lower balanced accuracy by 3–4 pp; on EfficientNet-B4 the share moves from 17% to 20%.

VI. LIMITATIONS

For a service never seen in training, separating lightly retouched faces from their originals stays at 70–73% balanced accuracy for every model we trained, below our target, and OURS reaches its sensitivity by sending 42% of that service’s originals to FILTER. Per-operation recall is below the published cross-vendor numbers for eye enlargement and lifting, and edit magnitudes could not be estimated. The test service applies one operation per image, so combined retouching is only tested on the training services. App filters are tested through four Celeb-DF-B presets, and OURS-lite accuses 40% of untreated Celeb-DF-B genuine frames. The evidence head localises part-level forgeries and filters, not whole-face swaps. OURS and OURS-lite are single training runs; two seeds of the CLIP model without part-level forgeries agree within 0.005 AUC.

VII. CONCLUSION

Beautification puts binary face-forgery detectors in a trade-off that no threshold resolves: on a paired corpus and on Celeb-DF-B, each of thirteen published detectors either accuses beautified genuine faces or lets beautified deepfakes through. A filter class removes the trade-off when it is trained with real forgeries, with degradation shared by all classes and with commercial renders split into single operations. A CLIP-based detector trained this way matches Effort and Forensics Adapter on cross-dataset forgery detection, is not measurably affected by beautification, and names the regions of part-level forgeries its decision depends on. Recognising light retouching from an unseen service is the part that remains open.

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